United International University

Trust Calibration in Legal AI:

Evaluating Interface Design and User Expertise in Detecting LLM Hallucinations for Bangladeshi Statutory Law

Supervised by Dr. Jannatun Noor Mukta

Presented to Dr. Riasat Azim

Sifatullah: 0112320240

Tasnuba Haque: 0112230833

Ibtida Ara: 0112320151

Aminul Islam Rifat: 0112320100

Ifrat-E-Sarwar Orpa: 0112320141

Can We Trust AI With Legal Advice?









The problem is not only AI errors.

The problem is whether humans can detect them.

If AI can confidently provide incorrect legal information, **how can users know when to trust it?**

Literature Review

Existing Research

- LLMs are increasingly used for legal question answering.
- RAG and citation-based systems help reduce hallucinations.
- Explainable AI (XAI) improves transparency of AI outputs.

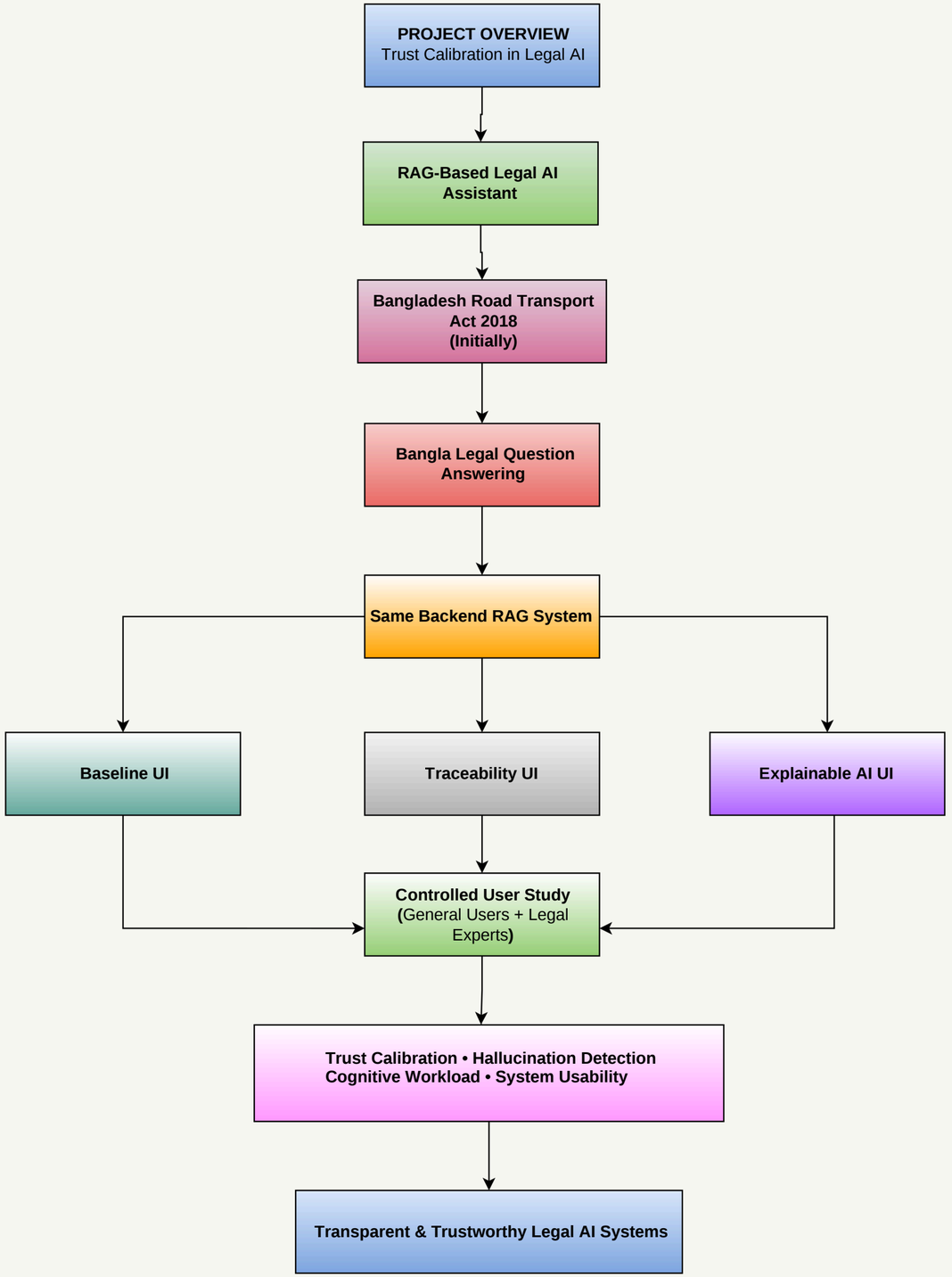
Research Gap

- AI systems can still hallucinate despite RAG.
- Limited research on Bangladeshi statutory law.
- Insufficient study of how users detect and respond to AI errors.
- Lack of trust calibration studies in Bangla legal contexts.

Our Research

- Develop a Bangla Legal AI Assistant using RAG.
- Design and compare 3 Human-AI interfaces.
- Study how interface design affects hallucination detection.
- Evaluate trust calibration among laymen and legal experts.

Project Overview



Motivation

The Core Problem & Domain Impact

1 **Legal AI Proliferation**

2 **High Risk of Hallucinations**



3 **Selected Focus: Bangladesh Road Transport Act, 2018(initially)**

Engineering Motivation

1

Backend (Pure RAG Architecture)

- Avoids expensive/unreliable full fine-tuning by grounding responses in official text.
- Challenge: RAG is not 100% error-free, especially in low-resource Bangla nuances.

2

Frontend (XAI & Trust Calibration)

- Focuses on the Human Factor: Helping users catch AI errors when absolute model accuracy isn't guaranteed.
- Research Gap: Lack of studies on how UI interventions (citations, uncertainty overlays) affect mental models.

Research Objectives



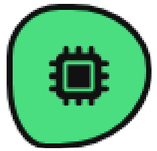
Statutory Dataset

Extract and process text from the Road Transport Act, 2018 to build the first Bangla benchmark dataset from Bangladeshi Statutory Law.



RAG Legal Assistant

Develop a Retrieval-Augmented Generation system to retrieve relevant laws and provide accurate answers, minimizing hallucinations.



Benchmark AI Models

Compare LLMs like GPT, Gemini, and BanglaBERT to evaluate their performance on Bengali legal queries and identify weaknesses.



Baseline Standard

Identify common AI errors and confusion points with local laws to establish a baseline and guide the custom training phase.

Research Objectives



Design & Compare UIs

Develop Simple, Citation-based, and Explainable AI (XAI) chatbots to evaluate how interfaces aid in verifying AI-generated answers.



Investigate User Trust

Conduct studies with ordinary users and experts to measure trust, workload, and error identification.



Provide Guidelines

Formulate practical recommendations from study findings to build more transparent, reliable, and trustworthy AI legal assistants.

Alignment with BAETE Complex Engineering Problem

P1

In-Depth Knowledge

HCI, XAI, RAG, NLP,
Legal Information Retrieval,
Web Development, Statistics

P2

Conflicting Requirements

- Transparency vs. Usability
- Explainability vs. Cognitive Load
- Trust vs. Over-reliance

P3

Depth of Analysis

- Interface designs
- User testing
- Trust & hallucination analysis

P6

Stakeholders

- Laymen
- Legal Experts
- Researchers

P7

Interdependence

Legal Documents → RAG System → User Interface
→ User Testing → Data Analysis

Alignment with BAETE Complex Engineering Problem

A1

Resources

• Legal Documents • AI Models • Vector Database
• Web Technologies • Legal Experts • User Participants

A2

Interaction

Integrating AI Accuracy, Transparency, and Usability

A3

Innovation

Combines RAG, XAI, and HCI to improve trust in Legal AI

A4

Societal Impact

Reduces risks of AI hallucinations in legal information

A5

Familiarity

Adapts existing HCI principles to the Bangladeshi Legal AI context

Thank You

Do you have any question? Feel free to ask.